Reg. No.: 22BCE(35)

Name :

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Continuous Assessment Test - II – October 2023					
Programme	:	B.Tech(CSE)	Semester	:	Fall 23-24
Course	:	Operating Systems	Code	:	BCSE303L
			ClassNbr	:	CH2023240100892, CH2023240100893, CH2023240101113
Faculty	:	Dr. Pravin Renold A, Dr. J. C. Kavitha, Dr. A. Menaka Pushpa	Slot	:	C2+TC2
Time	:	90 Minutes	Max. Marks	:	50

81.5

		Answer all Questions	Marks
1.		Suppose 3 business partners P1, P2, and P3 share a common bank account with deposit amount, withdrawal amount, and balance checking operations. Assume these three business partners initiate the money transaction operations concurrently. Under this circumstance, what issue will occur? Why does this problem occur? Explain it with a supporting code portion.	5
2.		At VIT University's library, multiple study rooms are available for students. To ensure that only one student can occupy a study room at a time and to manage access fairly, the university's SCOPE department has implemented an algorithm. Students arriving at the library take unique tickets to determine the order of access to the study rooms.	10
	(i)	Describe the problem that the university's SCOPE department is trying to solve and name the algorithm. **(2 Marks)**	
	(ii)	Consider the following four students (Student A, Student B, Student C, and Student D) who arrive at the library at different times and take tickets: Student A arrives first and takes ticket 1. Student B arrives next and takes ticket 2. Student C and D arrives at the same time, C takes ticket 4 and D takes ticket 3 Determine the order in which these students will be allowed to access the study rooms. Provide a step-by-step explanation of how the algorithm works in this scenario. **(4 Marks)**	
	(iii)	Write a pseudocode of the above-said algorithm. **(4 Marks)**	
3.		Consider a system where a counting semaphore is initialized to 3, controlling access to a critical section. Five processes P1, P2, P3, P4, and P5 sequentially request entry into this critical section. Consider the situation where the processes have varying priorities like P1-2, P2-3, P3-6, P4-1, P5-4	10

Satyaprakash Swain

		and the processes access the critical section based on highest value assigned as the priority.	
	(i)	Analyze which processes gain entry and which are placed in the waiting queue. **(2Marks)**	
	(ii)	Determine the value of the counting semaphore when P1, P2 and P3 is already executing in the critical section and while P2 exits P4 enters the critical section. **(3 Marks)**	
	(iii)	Write a pseudocode that not only implements the priority-based access but also handles the case where processes with equal priority request access simultaneously. Ensure that the pseudocode covers all possible scenarios and provides a fair and efficient access mechanism to the critical section. **(5 marks)**	
4.		Write a C program to implement the following game with 2 players using mutex–lock to ensure synchronization among players. Players move their coins on the board as per the random number generated by rolling two dice. If Player1 gets the even numbers in both dice then the turn will be given to Player2. Player2 moves the coin by one position on the left side. Similarly, Player2 gives a turn to Player1 if he gets the odd numbers in both dice. Player1 moves the coin on the right side for one position. This game has 10 rounds for each player. After the 10th round, the player who gets more chance to move the coin will be the winner.	10
5.		We operate a streaming platform that offers users a diverse selection of movies and TV series. Ensuring a seamless viewing experience is of utmost importance to us. To enhance the performance of our platform, You are asked to evaluate three distinct algorithms, labeled as 'i', 'ii' and 'iii' for the user's history of content selections: 'Action movie/Comedy Series/Documentary/Romance movie/Comedy series/Action movie/Thriller movie/Comedy series/Drama series/Action movie/Romance movie/Documentary/Crime series/Thriller movie/Romance movie/Action movie/Comedy Series/Documentary/Romance movie/Drama series'	15
	(i)	When a new movie/TV series needs to be loaded into the system's memory, it replaces the movie/TV series that has been in memory for the longest period of time. **(4 Marks)**	
	(ii)	When a new movie/TV series needs to be loaded into the system's memory, it replaces the movie/TV series that has experienced the least activity in recent history, with the assumption that it has become less pertinent for ongoing operations. **(4 Marks)**	
	(iii)	When a new movie or TV series is about to be loaded into the system's memory, it replaces the movie or TV series that is not expected to be accessed in the future. **(4 Marks)**	
	(iv)	Determine which algorithm is most preferred for the above customer preference. Justify your selection. **(3 Marks)**	
		Assume the memory slots of the device are free initially and the memory is divided into equal-sized frames of size 4.	

Satyaprakash Swain